Cart lab: change driving force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use controlled data to describe how net force affects acceleration.

SUCCESS CRITERIA - I can...

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “net force” to explain your model or result. Sentence starters are optional.

constant: not changing during the stated interval

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

01 Warm-up: bring back an earlier idea

A puck follows a circle at a steady 6 m/s. A student calls it equilibrium.
Draw velocity at two points and correct the claim.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

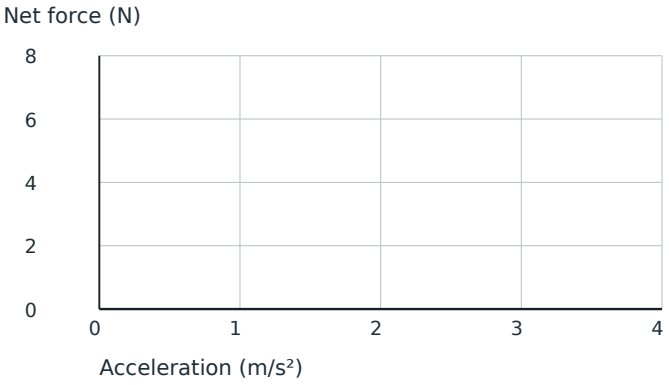
02 Worked example: explain the model

Reference: $a = F_{net}/M$; $F_{drive} = m_h g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

Worked reasoning: A 2 kg cart has net forces 2 and 6 N in two trials. Using a=F/m gives accelerations 1 and 3 m/s². Tripling net force triples acceleration.

Your turn: Draw a force-versus-acceleration graph for these trials.
Explain the meaning of the slope.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Interpret a force graph

For total mass 1.00 kg, use the ideal model table to find the slope of a versus net force.
State its units and recover the mass.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

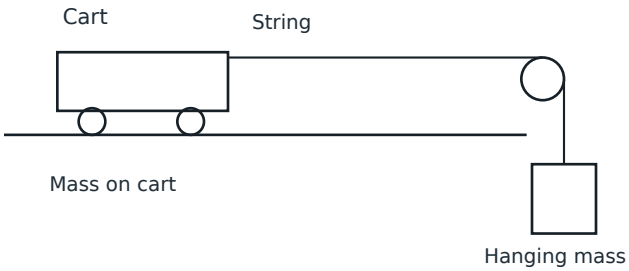
F_{net} (N)	a(m/s²)
0.245	0.245
0.490	0.490
0.735	0.735
0.980	0.980

Marzano check - circle: ① ② ③

04 Design the controlled trial

Use the setup model to explain why transferring a mass from cart to hanger changes the driving force but keeps total moving mass constant. Plan three force settings, with three acceleration trials at each setting.

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: ① ② ③

05 Both variables change

Two carts are tested under different conditions.
Use the table to calculate each acceleration. Why does tripling the net force not triple the acceleration in this comparison?

Sentence starter: Both … . They differ in … because … .

System	Net force (N)	Mass (kg)
A	2	1
B	6	2

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 0.3 kg system accelerates at 0.5 m/s².
Predict acceleration for 0.7 kg at the same net force.
Explain the effect of greater resistance.
Use the table to keep the controlled and changed quantities distinct.

Word help: acceleration = rate of change of velocity

Sentence starter: Both … . They differ in … because … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

07 Exit check

An a-versus-F_{net} graph has slope 2.0(m/s²)/N.
Find mass.
Predict a at 1.5 N.
Explain why plotting driving force instead of net force can produce an offset.

Sentence starter: I predict … . My model supports this because … .

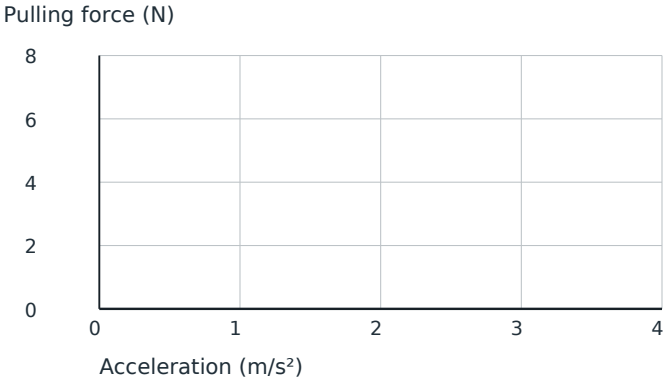
Marzano check - circle: ① ② ③

08 STRETCH: test the model

A 2 kg cart is pulled with 3, 5 and 7 N. Its accelerations are 1, 2 and 3 m/s². Plot pulling force versus acceleration.
Use the slope and intercept to infer a constant opposing force.

Word help: infer = use evidence to reach a conclusion

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: ① ② ③